

in ADAM T. GADDIS

SENIOR ENGINEER & TECH LEAD — DISTRIBUTED DATA PLATFORMS — ENGINEERING & IT LEADERSHIP

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🔗 Python, C#, Go, Rust

EXECUTIVE SUMMARY

Senior engineer and tech lead with 7+ years building distributed data systems, high-throughput ingestion pipelines, and customer-facing platforms across edge computing, energy, and finance. Fluent shipping in Go, Rust, Python, and C#, with production work in TypeScript and C++, ranging from petabyte-scale ingestion frameworks to 20 ms real-time control loops on Linux-based robotics controllers. Background bridges modern data platform engineering (Apache Iceberg, distributed query engines, streaming) with real-world IT operations and real-time IoT / SCADA ingestion, making me equally effective at systems engineering and the infrastructure work that keeps those systems running. Active open-source contributor to Apache Iceberg (Go) and WPILib. Tech lead, mentor, and nonprofit Co-Executive Director.

EXPERIENCE

• Cloudflare

Senior Data Engineer
Data Engineer

Dallas, TX (Remote)

Apr. 2025 – Present
May 2024 – Apr. 2025

- Core maintainer of Jetflow, Cloudflare's in-house Go-based data ingestion framework handling petabyte-scale pipelines, built around distributed systems primitives and highly concurrent workloads (see the public Cloudflare blog: [Building Jetflow](#)).
- Sole owner of Apache Iceberg support within Jetflow, enabling first-class Iceberg reads and writes across ingestion pipelines; active contributor to the open-source [apache/iceberg-go](#) project.
- Technical lead for the Data Infrastructure & Analytics team's migration off GCP (BigQuery and object storage) to Cloudflare R2, R2 Data Catalog, and a self-hosted query engine, projected to save \$1.5M+ annually.
- Leading the self-hosted query engine initiative (originating as a hackathon concept): integrating DuckDB as a Jetflow consumer reading from R2 Data Catalog and writing back to R2DC, and standing up StarRocks on Kubernetes as the distributed SQL layer.
- Selected to embed with Cloudflare's Billing Engineering (fintech) team for two quarters on the strength of my Interim Variable Billing work (\$40M quarterly recoup); architected and built their streaming metering and sinking platform in Go using Kafka and Iceberg, sustaining 100,000+ events per second. The system is becoming Cloudflare's unified billing pipeline, ultimately accounting for all revenue flowing through the company.
- Designed and built a customer-facing cost explorer (React front-end served by Cloudflare Workers and Containers) for one of Cloudflare's largest customers, marking the first successful POC of an application querying R2 Data Catalog directly to serve an end-user consumer.
- Owner of the Interim Variable Billing platform, a direct revenue-generating initiative recouping \$40M in quarterly revenue.
- Design and maintain Lighthouse, Cloudflare's internal sales and customer management platform, serving 1,000+ daily active users and unlocking hundreds of millions in annual contract value.
- Implemented Cloudflare's internal Retrieval-Augmented Generation (RAG) chatbot on the Lighthouse platform: built the ingestion pipeline loading hundreds of internal wikis into Cloudflare Vectorize and integrated it with Cloudflare Workers for AI-assisted internal search.
- Designed and implemented native data quality checks within the Jetflow framework using DuckDB, enabling efficient in-line validation during ingestion.
- Led the implementation of a customer metadata boundary across the BI data platform, enforcing consistent separation of customer-scoped metadata across ingestion, storage, and downstream consumers.
- Tech lead across multiple initiatives within Data Infrastructure & Analytics; deliver regular internal presentations (RAG implementation, open table formats, ClickHouse vs. Postgres, ingestion architecture, the migration from our legacy Spark-based ingestion framework, and the customer metadata boundary rollout) to audiences ranging from our sub-team to the 80+ member BI organization.

Technologies: Go, Rust, TypeScript, React, Python, Apache Iceberg, DuckDB, DataFusion, StarRocks, Kafka, Kubernetes, Docker, Cloudflare Workers, R2, R2 Data Catalog, Vectorize, Workers AI, Containers, BigQuery, ClickHouse

• **Adam Gaddis IT & Data Consulting**

Founder

Dallas, TX

April 2022 – Present

- Provide IT, data, and software consulting services to clients across the energy sector, spanning oil and gas production and real-time electrical metering / IoT.
- Helped multiple clients design and implement modern data infrastructure, automating ingestion and storage from key energy data providers such as Enverus, IHS, and ComboCurve.
- Designed and built a real-time electrical meter data platform: MQTT-based managed pub/sub firmware configuration delivered to field microcontrollers, streaming ingestion of meter and telemetry data into Apache Iceberg and Postgres, and user-facing Blazor applications on top of the platform.

Technologies: C# / Blazor, Apache Iceberg, Postgres, MQTT, Managed Pub/Sub, IoT / Embedded & Microcontrollers, Snowflake, Azure, GCP, Combocurve, PowerBI, Spotfire, Databricks, DBT, SSO, SCADA, SQL Server, Tailscale, Spark

• **Arch Energy Partners**

Head of Data & IT

Dallas, TX (Hybrid)

Mar. 2023 – May 2024

- As the founding Data / IT hire at a fast-moving oil and gas company, led the design and management of all software and data infrastructure, working autonomously to deliver scalable, resilient solutions.
- Built and operated the data ingestion platform in Go and Rust, orchestrated on GKE (Kubernetes) using Prefect for pod-based execution, integrating data from multiple providers (Enverus, RDS-based feeds, accounting, etc.) and processing hundreds of millions of daily well production records.
- Used Apache DataFusion for analytical compute across ingestion and transformation workloads, replacing a heavier JVM-based stack.
- Established and maintained a multi-cloud footprint across GCP, AWS, and Azure, optimizing for cost and performance; pioneered centralized warehousing on BigQuery.
- Built custom tooling to track and monitor active oil wells with real-time analytics, and designed the “Wells in Pay” Spotfire dashboard for Land and Accounting teams, providing immediate insight into well status and recovering missed payments.
- Centralized all transformation logic using Google Dataform to enforce consistency and transparency, and engineered multiple Google Cloud Functions to automate data transformation and storage workflows.
- Managed day-to-day IT operations (virtual machines, office VPN, cloud spend governance), eliminating reliance on third-party IT vendors and consultants by internalizing all infrastructure operations.
- Converted to an ongoing consulting engagement in May 2024 (see Adam Gaddis IT & Data Consulting above).

Technologies: Go, Rust, DataFusion, GKE / Kubernetes, Prefect, GCP, BigQuery, S3, GCS, Cloud Functions, SSO & OAuth, Spotfire, Geopandas, DBT / Dataform

• **Bank of America**

AVP Data Engineer II

Dallas, TX (Hybrid)

Sep. 2022 – Mar. 2023

- Overhauled real-time text-based segments of the data warehouse into a streaming Kafka + PySpark pipeline, replacing legacy SQL Server scripts and SSIS packages and achieving near-real-time processing of hundreds of millions of data points.
- Led the team’s SDLC modernization: migrated code to a centralized Bitbucket repository, documented ongoing systems, and stood up a CI/CD workflow using Bitbucket and Jenkins for seamless deployments.
- Drove the modernization strategy for moving off SQL Server, identifying and scoping the transition plan to Azure Synapse.
- Developed a Flask-based web application to revamp a report tracking system for a sister team, managing and updating over 1,000 internal reports encompassing more than 500,000 data points.

Technologies: Azure, Kafka, PySpark, Hadoop, MapReduce, HDFS, Sqoop, Impala, Azure Synapse, Bitbucket, Jenkins

• **LOOP**

Senior Data Engineer

Dallas, TX (Remote)

April 2022 – Sep. 2022

- Led key data engineering initiatives, focusing on building and managing robust data pipelines, remediating data governance issues, and centralizing data for company-wide analysis.
- Implemented and configured GCP-hosted Airflow (Cloud Composer), centralizing all pipeline scheduling into a unified orchestration system to improve data lineage and dependency management.
- Collaborated with a third-party data provider to develop Version 2 of our internal quoting model, deployed via Vertex AI & developed in C#, resulting in a 75% increase in the "Sign on First Quote" conversion rate.
- Resolved data discrepancies across vendors, from the policy administration system to the location tracking provider, by producing a fully approved, centralized data dictionary to enforce clarity and governance.
- Developed integration workflows to sync BigQuery data with Zendesk, fully automating the process of updating customer and lead records across sales and support platforms.

Technologies: Docker, GCP (Cloud Storage, Vertex AI, BigQuery, Cloud Composer), ERD, Data Governance

• Previous Experience Cont.

Pre: April 2022

*All roles predating 2022, condensed, see **LinkedIn** for full information on each of these roles* Jun. 2019 – Apr. 2022

- **Southwest Airlines** | *Data Platform Engineer Intern* | Jan. 2022 – April 2022 | Dallas, TX
Automated X12 EDI invoice ingestion using a custom Python parser, Airflow, and Postgres; supported ERP transformation planning and deployed pipelines via Docker and Jenkins.
- **Goldman Sachs** | *Data Platform Engineer Intern* | Jun. 2021 – Aug. 2021 | Dallas, TX
Built internal Python tools and libraries to streamline audit data workflows, improved test coverage from 70% to 91%, and reconciled on-prem vs. cloud schema mismatches for critical data systems.
- **University of North Texas at Dallas** | *Lead Analytics Engineer* | Aug. 2020 – Dec. 2021 | Dallas, TX
Led analytics and reporting efforts across Power BI, SAS, and Oracle; managed data requests, state reporting pipelines, and a team of five while ensuring full compliance with Texas education standards.
- **Scout Energy Partners** | *Data Platform Engineer* | Jun. 2019 – Aug. 2020 | Dallas, TX
Built and maintained real-time industrial data ingestion infrastructure (SCADA / IoT) for oil production analytics, alongside C# web apps, predictive models, Spotfire dashboards, Azure SQL Server development, and hybrid IT operations.

Technologies: Python, PyPI, Docker, Kubernetes, Azure, SQL Server, Oracle, Postgres, PowerBI, Tableau, Spotfire, Geopandas, Fiona, Shapely, Confluence, Jenkins

SKILLS

Languages: Go, Rust, Python, C#, TypeScript, C++, SQL, Scala, Dart

Distributed Data & Storage: Apache Iceberg, Kafka, MQTT, R2, R2 Data Catalog, BigQuery, ClickHouse, Snowflake, Databricks, Azure Synapse, Postgres, SQL Server, MongoDB, S3, GCS

Query & Compute Engines: DataFusion, DuckDB, StarRocks, Spark, DBT, Dataform

Orchestration: Airflow, Prefect, Azure Data Factory

Platforms: Cloudflare (Workers, R2, Vectorize, Workers AI, Containers), GCP, AWS, Azure

Systems & Infra: Linux, Kubernetes (GKE), Docker, CI/CD (Jenkins, GitLab, Bitbucket), FastAPI, Flask

IT & Infrastructure: SCADA, Cisco AnyConnect, Tailscale, Okta (SSO), Windows/Linux Server, On-Prem Systems

Frontend & App: React, TypeScript, Flutter / Dart, Blazor

Analytics & ML: PowerBI, Spotfire, Vertex AI, Workers AI, Vectorize, Keras, OpenCV

Governance & Compliance: Monte Carlo, Great Expectations, dbdiagram.io

EDUCATION

University of North Texas at Dallas

Dallas, TX

M.B.A Strategic Management Concentration — GPA: 4.0/4.0

Aug. 2024 – July. 2025

University of North Texas at Dallas

Dallas, TX

B.B.A in Analytics — GPA: 3.7/4.0

Aug. 2020 – Dec. 2022

Dallas College

Dallas, TX

A.A.S in Software Development — GPA: 3.6/4.0.

Aug. 2016 – May. 2020

LEADERSHIP, BOARD, ADVISORY & COMMUNITY ENGAGEMENT

Proven operational leadership in nonprofit STEM education with budgetary control, executive stakeholder engagement, and governance alignment. Experienced in building infrastructure, compliance systems, and educational pipelines, well-positioned for board and advisory contributions in tech-forward organizations.

FIRST Robotics Competition Team 3005

Dallas, TX

Co-executive Director & Lead Software Mentor

2020 – Present

- Lead mentor and administrator for FRC Team 3005 and FTC teams at Emmett J. Conrad High School in Dallas ISD, serving students in the Vickery Meadows area, a designated refugee community with highly diverse and underserved demographics.
- Mentor students in systems-level C++ programming on Linux-based roboRIO controllers, implementing 20 ms watchdog control loops, real-time sensor fusion, and vision-based field localization for 130 lb competition robots operating at 5+ m/s. Performance-critical C++ code is exposed to student developers via pybind11 Python bindings.
- Active contributor to [WPILib](#) ([wpilibsuite/allwpilib](#)), the open-source C++/Java systems framework that powers all FRC robots.
- Mentor students building the team's competition scouting and analytics application in Flutter / Dart, used live during match play to inform strategy and alliance selection.
- Contributed 1,000+ hours annually mentoring students and managing team operations, including software education, IT infrastructure, and team logistics.
- Oversee cloud infrastructure and IT systems, leveraging nonprofit credits and mentoring students in the development of scouting, data collection, and analytics apps used during competition.
- Assumed full leadership of the program in 2025 alongside my wife, taking responsibility for administration, budgeting, grant writing, and maintaining relationships with school leadership and key sponsors such as the Vickery Meadows Youth Foundation.
- Manage an annual nonprofit budget of \$60,000–\$80,000 to fund robotics competitions, equipment, travel, and outreach, ensuring long-term sustainability and community impact.
- Lead all software and IT initiatives, including robot programming, while fostering student-driven development in a hands-on, collaborative learning environment.
- Mentor approximately 30 students annually and support competitive excellence, including back-to-back Texas UIL State Championship titles.